

TOI-1811b

R-2332

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TOI1811_240127 · created 2026-08-18T23:24:47+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460336.8289 +/- 0.0036 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1659 +/- 0.0054
Transit depth (Rp/Rs)^2	2.75 +/- 0.18 [%]
Orbital Inclination (inc)	86.8 +/- 0.56
Airmass coefficient 1 (a1)	0.985 +/- 0.014
Airmass coefficient 2 (a2)	0.0074 +/- 0.0048
Scatter in the residuals of the lightcurve fit is	1.47 %
Best Comparison Star	#3 - [314, 390]
Optimal Method	PSF photometry
Transit Duration (day)	0.0894 +/- 0.0081

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.1659 ± 0.0054 vs 0.13272 (deviation 5.99σ)
Duration fit vs published	0.0894 d vs 0.0813 d (deviation 0.98σ)
Mid-time O–C	1.0 min
Depth SNR	30.0
Residual scatter	1.47 %

Light curve

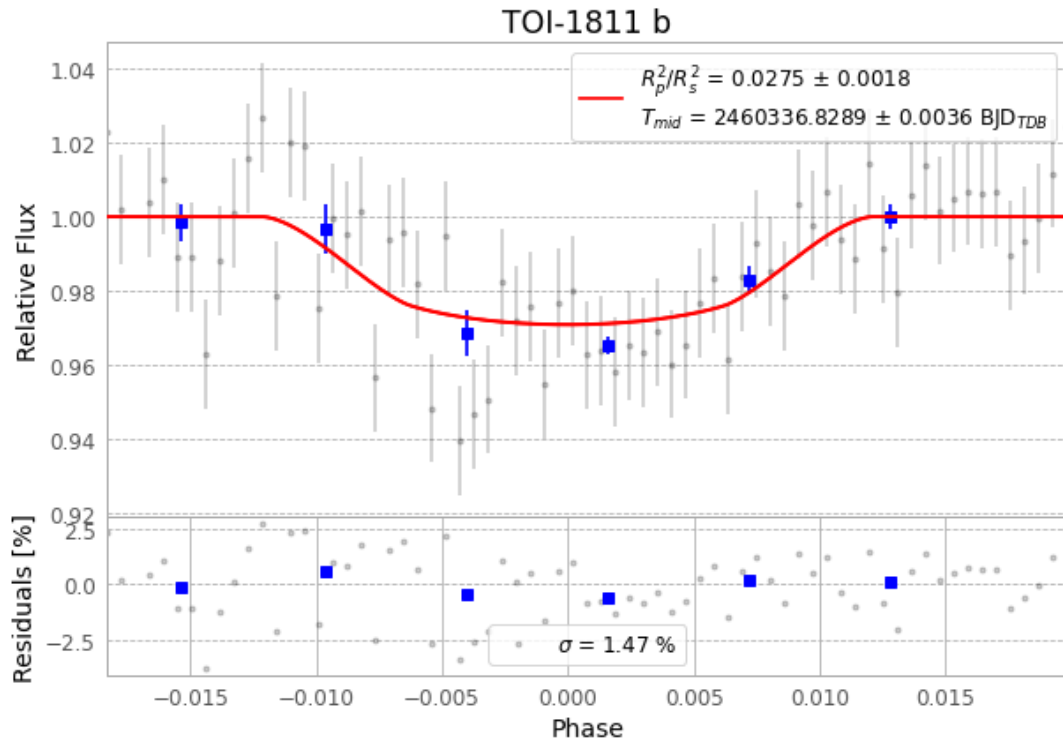


Figure 1 — FinalLightCurve_TOI-1811 b_2024-01-26.png (SHA-256 b15da462ce16ac2a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [407, 255], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 19212685a98177be...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

69 FITS frames (45 MB), TOI1811240127060917.FITS → TOI1811240127093315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI1811-240127.log (d115a9ec5446...), FinalLightCurve_TOI-1811 b_2024-01-26.png (b15da462ce16...), FinalLightCurve_TOI-1811 b_2024-01-26.csv (f15a58fd939b...)

Compute resources

Wall time 108.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-18 23:24Z.